

SPSS Assignment 2

Continuous (Scale) Data Analysis: Descriptives, T-Tests, ANOVA, Repeated Measures, Correlation, Regression, and Factor/Reliability Analysis

Datasets: Dataset 1 – Student_Academic_Performance.sav (N = 200); Dataset 2 – Employee_Job_Satisfaction.sav (N = 200)

Significance level: $\alpha = .05$ for all tests unless otherwise stated

Section A. Descriptive Statistics & Visualization of Continuous Variables

Q1. Descriptive Statistics – Age, Study_Hours_Per_Week, Attendance_Percentage, Final_Exam_Score (Dataset 1)

SPSS Steps

- Analyze → Descriptive Statistics → Frequencies
- Move Age, Study_Hours_Per_Week, Attendance_Percentage, Final_Exam_Score into the Variable(s) box
- Uncheck “Display frequency tables”
- Statistics → check Mean, Median, Mode, Std. deviation, Range → Continue → OK

Output – Summary Table

Variable	N	Mean	Median	Mode	SD	Range
Age	200	21.04	21.00	21.0	1.71	9.0
Study Hours/Week	200	12.18	12.15	12.0	3.75	21.0
Attendance (%)	200	79.52	79.70	100.0	10.39	47.8
Final Exam Score	200	67.03	67.45	59.8	11.17	93.8

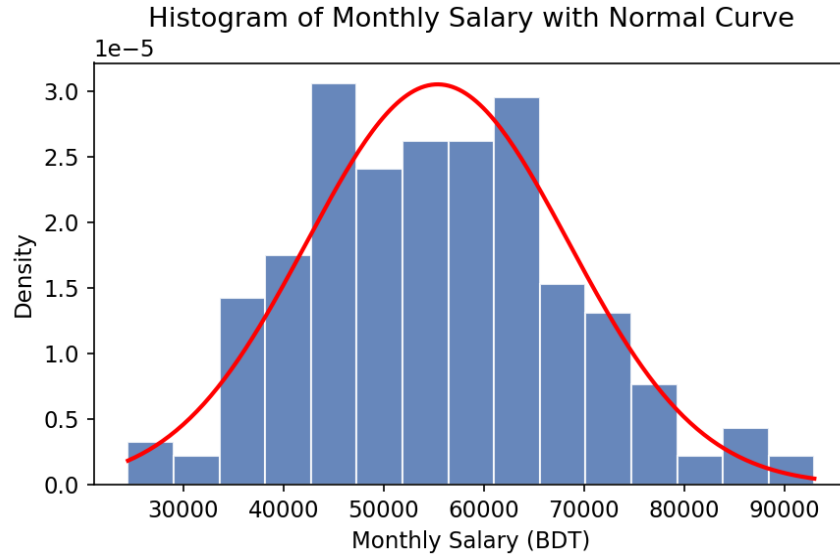
APA-Style Interpretation

Students were, on average, 21.04 years old (SD = 1.71, range = 9), studied 12.18 hours per week (SD = 3.75, range = 21.0), had a mean attendance of 79.52% (SD = 10.39, range = 47.8), and scored 67.03 on the final exam on average (SD = 11.17, range = 93.8). The wide range for Final_Exam_Score (5.7–99.5) suggests some very low-performing outliers, which is examined further in Section B.

Q2. Histogram of Monthly_Salary and Boxplot of Performance_Rating by Department (Dataset 2)

SPSS Steps

- Graphs → Legacy Dialogs → Histogram → Variable: Monthly_Salary → check “Display normal curve” → OK
- Graphs → Legacy Dialogs → Boxplot → Simple, Summaries for groups of cases → Variable: Performance_Rating; Category axis: Department → OK



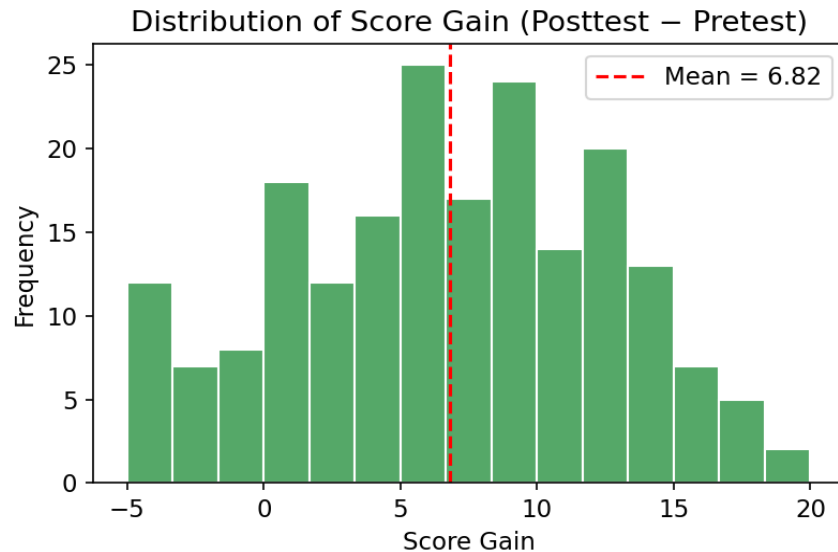
APA-Style Interpretation

Monthly_Salary ($M = \text{BDT } 55,383$, $SD = \text{BDT } 13,063$) showed mild positive skewness ($\text{skewness} = 0.33$) and slightly platykurtic shape ($\text{kurtosis} = -0.09$), both well within the ± 1 guideline for approximate normality, so no major departure from normality was evident in the histogram, which closely tracks the overlaid normal curve. The boxplot of Performance_Rating by department shows broadly similar medians across departments (ranging from about 61.7 in IT to 66.8 in HR), with a small number of mild outliers: one low value in HR (41.7), one high value in Operations (97.8), and two outliers in Sales (a high value of 95.3 and a low value of 36.9). These are isolated cases rather than a systematic pattern and do not appear to distort the overall group comparisons.

Q3. Compute $\text{Score_Gain} = \text{Posttest_Score} - \text{Pretest_Score}$ (Dataset 1)

SPSS Steps

- Transform → Compute Variable
- Target Variable: Score_Gain
- Numeric Expression: Posttest_Score – Pretest_Score → OK
- Analyze → Descriptive Statistics → Descriptives → Score_Gain → OK



Output

N	Mean	Median	SD	Min	Max
200	6.82	7.05	5.82	-5.0	20.0

APA-Style Interpretation

On average, students improved by **6.82** points ($SD = 5.82$) from pretest to posttest, a gain that a one-sample t-test confirmed was significantly different from zero, $t(199) = 16.58$, $p < .001$. Although a small number of students (with negative Score_Gain values, minimum = -5.0) performed worse on the posttest, the large majority showed improvement (maximum gain = 20.0 points), indicating that, overall, the intervention/instruction between the two tests was associated with a meaningful increase in performance.

Section B. Normality Check & Outlier Detection (Z-Scores)

Q4. Normality of Final_Exam_Score – Shapiro-Wilk Test & Q-Q Plot (Dataset 1)

SPSS Steps

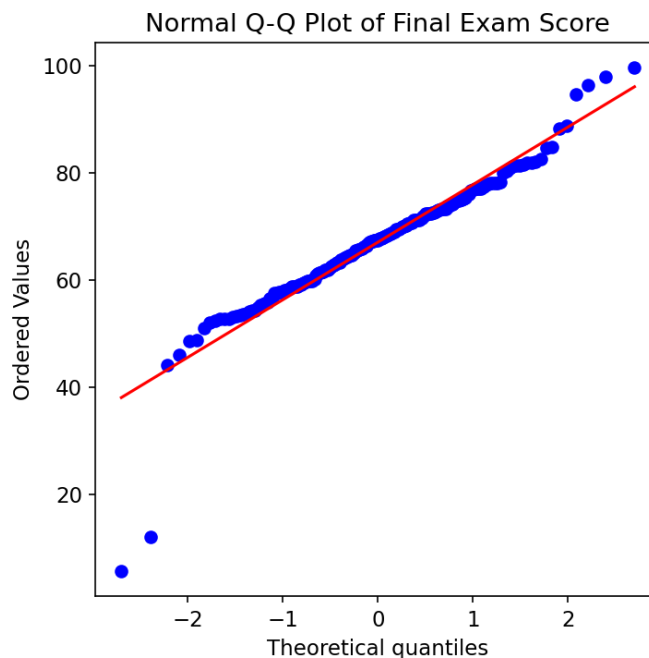
- Analyze → Descriptive Statistics → Explore
- Dependent List: Final_Exam_Score
- Plots → check Normality plots with tests → Continue → OK

H₀ (null): Final_Exam_Score is normally distributed in the population.

H₁ (alternative): Final_Exam_Score is not normally distributed in the population.

Output – Tests of Normality

Test	Statistic	df	p
Shapiro-Wilk	.915	200	< .001
Skewness		Kurtosis	
-1.075		6.740	



APA-Style Interpretation

The Shapiro-Wilk test indicated that Final_Exam_Score significantly deviated from a normal distribution, $W(200) = .915$, $p < .001$. This is corroborated by the strong negative skew (-1.075) and high kurtosis (6.740), and by the Q-Q plot, in which points depart noticeably from the diagonal reference line in the lower tail — consistent with the small cluster of very low exam scores (as few as 5.7) identified earlier.

Decision: Reject H_0 ($p < .05$) — the normality assumption is not met for Final_Exam_Score. Subsequent parametric tests using this variable should be interpreted with this violation in mind, although t-tests and ANOVA are considered fairly robust to non-normality with a sample size of $N = 200$.

Q5. Standardized Z-Scores for Monthly_Salary & Outlier Identification (Dataset 2)

SPSS Steps

- Analyze → Descriptive Statistics → Descriptives
- Variable: Monthly_Salary → check “Save standardized values as variables” → OK (creates ZMonthly_Salary)
- Data → Select Cases (or Data → Sort/Filter) with condition $ABS(ZMonthly_Salary) > 3$ to flag outliers

Output

N	Mean (BDT)	SD (BDT)	Min Z	Max Z
200	55,383.07	13,063.01	-2.37	2.88

APA-Style Interpretation

Standardized Z-scores for Monthly_Salary ranged from -2.37 to 2.88. No employee had an absolute Z-score greater than 3 (0 of 200 cases), indicating that there were no extreme univariate outliers in Monthly_Salary using the conventional $|Z| > 3$ criterion. The salary distribution, while slightly positively skewed (see Q2), does not contain any values extreme enough to warrant removal or winsorizing prior to further analysis.

Section C. T-Tests

Q6. One-Sample T-Test: Attendance_Percentage vs. Test Value = 75 (Dataset 1)

SPSS Steps

- Analyze → Compare Means → One-Sample T Test
- Test Variable(s): Attendance_Percentage; Test Value: 75 → OK

H₀ (null): μ (mean Attendance_Percentage) = 75.

H₁ (alternative): μ (mean Attendance_Percentage) $\neq$ 75.

Output – One-Sample Statistics

N	Mean	SD	SE Mean
200	79.53	10.39	.735

One-Sample Test (Test Value = 75)

t	df	p (2-tailed)	Mean Diff.	95% CI of the Difference
6.159	199	< .001	4.525	[3.076, 5.973]

APA-Style Interpretation

A one-sample t-test showed that mean attendance ($M = 79.53\%$, $SD = 10.39$) was significantly higher than the hypothesized value of 75%, $t(199) = 6.16$, $p < .001$, 95% CI [3.08, 5.97], $d = 0.44$ (a small-to-medium effect). Students in this sample attended class, on average, about 4.5 percentage points more often than the benchmark of 75%.

Decision: Reject H_0 ($p < .05$).

Q7. Independent-Samples T-Test: Final_Exam_Score by Gender (Dataset 1)

SPSS Steps

- Analyze → Compare Means → Independent-Samples T Test
- Test Variable: Final_Exam_Score; Grouping Variable: Gender (define groups 1, 2) → OK

H₀ (null): $\mu(\text{Male}) = \mu(\text{Female})$ for Final_Exam_Score.

H₁ (alternative): $\mu(\text{Male}) \neq \mu(\text{Female})$ for Final_Exam_Score.

Output – Group Statistics

Gender	N	Mean	SD	SE Mean
Male	100	65.83	12.44	1.243
Female	100	68.24	9.65	.965

Independent Samples Test

	Levene's F	Levene's p	t	df	p (2-tailed)	Mean Diff.	95% CI
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Equal variances assumed	.853	.357	−1.536	198	.126	−2.417	[−5.52, 0.69]
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APA-Style Interpretation

Levene's test indicated that the assumption of homogeneity of variance was met, $F = 0.85$, $p = .357$, so the “equal variances assumed” row was used. An independent-samples t-test showed no statistically significant difference in Final_Exam_Score between male ($M = 65.83$, $SD = 12.44$) and female ($M = 68.24$, $SD = 9.65$) students, $t(198) = -1.54$, $p = .126$, 95% CI for the mean difference $[-5.52, 0.69]$, $d = -0.22$ (small effect).

Decision: Fail to reject H_0 ($p > .05$).

Q8. Paired-Samples T-Test: Q1_Engagement vs. Q4_Engagement (Dataset 2)

SPSS Steps

- Analyze → Compare Means → Paired-Samples T Test
- Pair 1: Q1_Engagement – Q4_Engagement → OK

H_0 (null): $\mu(Q1_Engagement) = \mu(Q4_Engagement)$ (no change in engagement from Quarter 1 to Quarter 4).

H_1 (alternative): $\mu(Q1_Engagement) \neq \mu(Q4_Engagement)$ (engagement changed from Quarter 1 to Quarter 4).

Output – Paired Samples Statistics

	Mean	N	SD
Q1_Engagement	54.40	200	11.88
Q4_Engagement	61.90	200	13.75

Paired Samples Test

Mean Diff. (Q1–Q4)	SD	t	df	p (2-tailed)	95% CI
−7.50	6.92	−15.336	199	< .001	[−8.46, −6.54]

APA-Style Interpretation

There was a statistically significant increase in engagement from Quarter 1 ($M = 54.40$, $SD = 11.88$) to Quarter 4 ($M = 61.90$, $SD = 13.75$), $t(199) = -15.34$, $p < .001$, 95% CI for the mean difference $[-8.46, -6.54]$. The effect size was very large, Cohen's $d_z = 1.08$, indicating a substantial and consistent rise in employee engagement across the year (r between the two occasions = .86).

Decision: Reject H_0 ($p < .05$).

Section D. One-Way ANOVA

Q9. Final_Exam_Score across Department, with Tukey HSD Post-Hoc (Dataset 1)

SPSS Steps

- Analyze → Compare Means → One-Way ANOVA
- Dependent List: Final_Exam_Score; Factor: Department
- Options → Homogeneity of variance test → Continue
- Post Hoc → Tukey → Continue → OK

H₀ (null): The mean Final_Exam_Score is equal across all four Department groups ($\mu_1 = \mu_2 = \mu_3 = \mu_4$).

H₁ (alternative): At least one Department group has a different mean Final_Exam_Score.

Descriptives

Department	n	Mean	SD
Engineering	52	71.22	9.68
Science	47	69.99	7.24
Business	61	65.08	12.61
Arts	40	61.11	11.46

Test of Homogeneity of Variance (Levene)

Levene Statistic	df1	df2	p
1.879	3	196	.135

Levene's test was not significant ($p = .135$), so the assumption of homogeneity of variance was satisfied and the standard ANOVA F-test (and Tukey HSD) is appropriate.

ANOVA

Source	SS	df	MS	F	p
Between Groups	2959.16	3	986.39	8.845	< .001
Within Groups	21858.18	196	111.52		
Total	24817.33	199			

η^2 (eta squared) = .119, a medium-to-large effect.

Tukey HSD Post-Hoc Comparisons

Comparison	Mean Diff.	p (adj.)	Sig.?
Arts vs. Business	3.97	.254	No
Arts vs. Engineering	10.11	< .001	Yes
Arts vs. Science	8.88	.001	Yes
Business vs. Engineering	6.14	.013	Yes
Business vs. Science	4.91	.082	No
Engineering vs. Science	-1.24	.938	No

APA-Style Interpretation

A one-way ANOVA revealed a statistically significant effect of Department on Final_Exam_Score, $F(3, 196) = 8.85$, $p < .001$, $\eta^2 = .119$. Tukey HSD post-hoc comparisons showed that Engineering ($M = 71.22$) and Science ($M = 69.99$) students scored significantly higher than Arts students ($M = 61.11$; both $p \leq .001$), and Engineering students also scored significantly higher than Business students ($M = 65.08$, $p = .013$). No other pairwise comparisons were significant (all $p > .05$), including Business vs. Arts, Business vs. Science, and Engineering vs. Science.

Decision: Reject H_0 ($p < .05$).

Q10. Monthly_Salary by Education_Level, with Bonferroni Post-Hoc (Dataset 2)

SPSS Steps

- Analyze → Compare Means → One-Way ANOVA
- Dependent List: Monthly_Salary; Factor: Education_Level
- Options → Homogeneity of variance test → Continue
- Post Hoc → Bonferroni → Continue → OK

H₀ (null): The mean Monthly_Salary is equal across all Education_Level groups ($\mu_1 = \mu_2 = \mu_3$).

H₁ (alternative): At least one Education_Level group has a different mean Monthly_Salary.

Descriptives

Education Level	n	Mean (BDT)	SD (BDT)
Bachelor	104	52,017.10	11,903.66
Master	84	57,633.12	12,360.01
PhD	12	68,804.42	16,287.12

Test of Homogeneity of Variance (Levene)

Levene Statistic	df1	df2	p
0.478	2	197	.621

Levene's test was not significant ($p = .621$); the assumption of equal variances is satisfied.

ANOVA

Source	SS	df	MS	F	p
Between Groups	3,765,156,000	2	1,882,578,000	12.283	< .001
Within Groups	30,192,670,000	197	153,262,300		
Total	33,957,826,000	199			

η^2 (eta squared) = .111, a medium-to-large effect.

Bonferroni Post-Hoc Comparisons

Comparison	Mean Diff. (BDT)	p (adj.)	Sig.?
Bachelor vs. Master	-5,616.02	.006	Yes
Bachelor vs. PhD	-16,787.32	< .001	Yes
Master vs. PhD	-11,171.30	.018	Yes

APA-Style Interpretation

A one-way ANOVA showed a statistically significant effect of Education_Level on Monthly_Salary, $F(2, 197) = 12.28$, $p < .001$, $\eta^2 = .111$. Bonferroni-adjusted pairwise comparisons indicated that all three groups differed significantly from one another: PhD holders ($M = \text{BDT } 68,804$) earned significantly more than both Master's ($M = \text{BDT } 57,633$, $p = .018$) and Bachelor's degree holders ($M = \text{BDT } 52,017$, $p < .001$), and Master's degree holders earned significantly more than Bachelor's degree holders ($p = .006$). Salary therefore increased monotonically with level of education.

Decision: Reject H_0 ($p < .05$).

Section E. Repeated Measures ANOVA

Q11. Engagement across Four Quarters: Q1_Engagement – Q4_Engagement (Dataset 2)

SPSS Steps

- Analyze → General Linear Model → Repeated Measures
- Within-Subject Factor Name: Quarter; Number of Levels: 4 → Add → Define
- Assign Q1_Engagement, Q2_Engagement, Q3_Engagement, Q4_Engagement to the four levels
- Options → check Descriptive statistics, Estimates of effect size → Continue
- Post Hoc / pairwise comparisons → Bonferroni adjustment → OK

H₀ (null): Mean engagement is equal across all four quarters ($\mu_1 = \mu_2 = \mu_3 = \mu_4$).

H₁ (alternative): Mean engagement differs across at least one pair of quarters.

Descriptive Statistics

Quarter	Mean	SD	N
Q1_Engagement	54.40	11.88	200
Q2_Engagement	56.69	12.71	200
Q3_Engagement	59.87	12.80	200
Q4_Engagement	61.90	13.75	200

Mauchly's Test of Sphericity

Mauchly's W	Approx. χ^2	df	p	Greenhouse-Geisser ϵ
0.404	179.04	5	< .001	.641

Mauchly's test indicated that the assumption of sphericity had been violated, $\chi^2(5) = 179.04$, $p < .001$, so the Greenhouse-Geisser correction ($\epsilon = .641$) was applied to the degrees of freedom of the within-subjects effect.

Tests of Within-Subjects Effects (Greenhouse-Geisser Corrected)

Source	SS	df (corrected)	MS	F	p	Partial η^2
Quarter	6634.67	1.92	3453.68	164.81	< .001	.453
Error(Quarter)	8011.17	382.63	20.94			

Pairwise Comparisons (Bonferroni-Adjusted)

Comparison	Mean Diff.	t	p (adj.)	Sig.?
Q1 vs. Q2	-2.30	-7.91	< .001	Yes
Q1 vs. Q3	-5.47	-13.93	< .001	Yes
Q1 vs. Q4	-7.50	-15.34	< .001	Yes
Q2 vs. Q3	-3.17	-11.60	< .001	Yes
Q2 vs. Q4	-5.20	-12.49	< .001	Yes
Q3 vs. Q4	-2.03	-7.21	< .001	Yes

APA-Style Interpretation

Mauchly's test indicated that the assumption of sphericity was violated, $\chi^2(5) = 179.04$, $p < .001$, so degrees of freedom were corrected using the Greenhouse-Geisser estimate ($\epsilon = .641$). Engagement scores changed significantly across the four quarters, $F(1.92, 382.63) = 164.81$, $p < .001$, partial $\eta^2 = .453$ (a very large effect). Bonferroni-adjusted pairwise comparisons showed that engagement increased significantly at every successive quarter ($Q1 = 54.40 \rightarrow Q2 = 56.69 \rightarrow Q3 = 59.87 \rightarrow Q4 = 61.90$, all pairwise $p < .001$), indicating a steady, monotonic rise in employee engagement across the year rather than a one-off jump between any two particular quarters.

Decision: Reject H_0 ($p < .05$).

Section F. Correlation Analysis

Q12. Pearson Correlation Matrix: Study_Hours_Per_Week, Attendance_Percentage, Final_Exam_Score (Dataset 1)

SPSS Steps

- Analyze $\rightarrow$ Correlate $\rightarrow$ Bivariate
- Variables: Study_Hours_Per_Week, Attendance_Percentage, Final_Exam_Score
- Correlation Coefficients: Pearson; Test of Significance: Two-tailed $\rightarrow$ OK

H_0 (null): There is no linear relationship between each pair of variables ($\rho = 0$).

H₁ (alternative): There is a linear relationship between each pair of variables ($\rho \neq 0$).

Output – Correlation Matrix (N = 200)

	Study Hours	Attendance %	Final Exam Score
Study Hours	1.000	$r = .002, p = .977$	$r = .345, p < .001$
Attendance %		1.000	$r = .274, p < .001$
Final Exam Score			1.000

APA-Style Interpretation

Study_Hours_Per_Week was moderately, positively correlated with Final_Exam_Score, $r(198) = .345, p < .001$, indicating that students who studied more hours per week tended to score higher on the final exam. Attendance_Percentage also showed a small-to-moderate positive correlation with Final_Exam_Score, $r(198) = .274, p < .001$. However, Study_Hours_Per_Week and Attendance_Percentage were essentially uncorrelated with each other, $r(198) = .002, p = .977$, suggesting that studying more and attending more classes are largely independent behaviors among these students, each contributing separately to exam performance.

Decision: Reject H_0 for Study Hours–Final Exam and Attendance–Final Exam (both $p < .05$); fail to reject H_0 for Study Hours–Attendance ($p > .05$).

Q13. Spearman Correlation: Experience_Years and Performance_Rating (Dataset 2)

SPSS Steps

- Analyze → Correlate → Bivariate
- Variables: Experience_Years, Performance_Rating
- Correlation Coefficients: Spearman; Test of Significance: Two-tailed → OK

H₀ (null): There is no monotonic relationship between Experience_Years and Performance_Rating ($\rho_s = 0$).

H₁ (alternative): There is a monotonic relationship between Experience_Years and Performance_Rating ($\rho_s \neq 0$).

Output

Spearman's rho	p (2-tailed)	N
.160	.023	200

APA-Style Interpretation

A Spearman correlation revealed a statistically significant, but weak, positive association between years of work experience and supervisor-rated performance, $r_s(198) = .160, p = .023$. More experienced employees tended to receive slightly higher performance ratings, although the relationship is weak and experience alone explains only a small proportion of the variation in performance ($r_s^2 \approx 2.6\%$).

Decision: Reject H_0 ($p < .05$).

Section G. Simple & Multiple Linear Regression

Q14. Simple Linear Regression: Final_Exam_Score on Study_Hours_Per_Week (Dataset 1)

SPSS Steps

- Analyze → Regression → Linear
- Dependent: Final_Exam_Score; Independent(s): Study_Hours_Per_Week → OK

H₀ (null): Study_Hours_Per_Week does not significantly predict Final_Exam_Score ($\beta = 0$).

H₁ (alternative): Study_Hours_Per_Week significantly predicts Final_Exam_Score ($\beta \neq 0$).

Model Summary

R	R ²	Adj. R ²	F (1, 198)	p
.345	.119	.115	26.80	< .001

Coefficients

Predictor	B	SE	β (standardized)	t	p
Constant	54.526	2.528		21.57	< .001
Study_Hours_Per_Week	1.027	.198	.345	5.18	< .001

APA-Style Interpretation

A simple linear regression showed that Study_Hours_Per_Week significantly predicted Final_Exam_Score, $F(1, 198) = 26.80$, $p < .001$, accounting for 11.9% of the variance ($R^2 = .119$, adjusted $R^2 = .115$). The regression equation was: $\text{Final_Exam_Score} = 54.53 + 1.03 \times \text{Study_Hours_Per_Week}$. Each additional hour of weekly study was associated with an increase of approximately 1.03 points on the final exam ($\beta = .345$, $t = 5.18$, $p < .001$).

Decision: Reject H₀ ($p < .05$).

Q15. Multiple Linear Regression: Final_Exam_Score on Study_Hours_Per_Week, Attendance_Percentage, Stress_Level (Dataset 1)

SPSS Steps

- Analyze → Regression → Linear
- Dependent: Final_Exam_Score; Independent(s): Study_Hours_Per_Week, Attendance_Percentage, Stress_Level; Method: Enter
- Statistics → check Collinearity diagnostics → Continue → OK

H₀ (null): None of Study_Hours_Per_Week, Attendance_Percentage, or Stress_Level predicts Final_Exam_Score (all $\beta = 0$).

H₁ (alternative): At least one of Study_Hours_Per_Week, Attendance_Percentage, or Stress_Level predicts Final_Exam_Score (at least one $\beta \neq 0$).

Model Summary

R	R ²	Adj. R ²	F (3, 196)	p
.493	.243	.231	20.95	< .001

Coefficients

Predictor	B	SE	t	p	VIF
Constant	40.495	6.365	6.36	< .001	
Study_Hours_Per_Week	1.034	.185	5.59	< .001	1.000
Attendance_Percentage	.265	.067	3.94	< .001	1.014
Stress_Level	-1.323	.371	-3.56	< .001	1.015

APA-Style Interpretation

The three-predictor model significantly predicted *Final_Exam_Score*, $F(3, 196) = 20.95$, $p < .001$, $R^2 = .243$, adjusted $R^2 = .231$ — an improvement over the simple regression in Q14. All three predictors were statistically significant: *Study_Hours_Per_Week* ($B = 1.034$, $p < .001$) and *Attendance_Percentage* ($B = 0.265$, $p < .001$) were positively associated with exam performance, while *Stress_Level* was negatively associated ($B = -1.323$, $p < .001$), such that each one-point increase in self-reported stress was associated with roughly a 1.3-point decrease in final exam score, holding the other predictors constant. All VIF values were close to 1.0 (range 1.000–1.015), indicating no multicollinearity concern among the predictors.

Decision: Reject H_0 ($p < .05$) — all three predictors are significant.

Q16. Multiple Linear Regression: *Monthly_Salary* on *Experience_Years*, *Performance_Rating*, *Age* (Dataset 2)

SPSS Steps

- Analyze → Regression → Linear
- Dependent: *Monthly_Salary*; Independent(s): *Experience_Years*, *Performance_Rating*, *Age*; Method: Enter
- Statistics → check Collinearity diagnostics → Continue → OK

H_0 (null): None of *Experience_Years*, *Performance_Rating*, or *Age* predicts *Monthly_Salary* (all $\beta = 0$).

H_1 (alternative): At least one of *Experience_Years*, *Performance_Rating*, or *Age* predicts *Monthly_Salary* (at least one $\beta \neq 0$).

Model Summary

R	R ²	Adj. R ²	F (3, 196)	p
.808	.652	.646	122.3	< .001

Coefficients

Predictor	B	SE	t	p	VIF
Constant	31,470	4622.4	6.81	< .001	
<i>Experience_Years</i>	2,047.54	114.9	17.81	< .001	1.115
<i>Performance_Rating</i>	39.41	55.6	0.71	.479	1.050
<i>Age</i>	60.52	87.2	0.69	.488	1.067

Collinearity / Multicollinearity Check

All VIF values (1.05–1.12) are far below the conventional threshold of 10 (and below the more conservative threshold of 5), indicating that multicollinearity among *Experience_Years*, *Performance_Rating*, and *Age* is not a concern in this model.

APA-Style Interpretation

The overall model was highly significant, $F(3, 196) = 122.30$, $p < .001$, and explained 65.2% of the variance in *Monthly_Salary* ($R^2 = .652$, adjusted $R^2 = .646$) — a large effect. *Experience_Years* was the only statistically significant predictor, $B = 2,047.54$, $t = 17.81$, $p < .001$, indicating that each additional year of work experience was associated with an increase of roughly BDT 2,048 in monthly salary, holding performance and age constant. Neither *Performance_Rating* ($p = .479$) nor *Age* ($p = .488$) contributed significantly to the model once *Experience_Years* was accounted for, suggesting that in this organization, salary is driven predominantly by tenure/experience rather than by current performance ratings or age.

Decision: Reject H_0 for the overall model and for Experience_Years ($p < .05$); fail to reject H_0 for Performance_Rating and Age ($p > .05$).

Section H. Factor Analysis & Reliability Analysis (Dataset 2)

Q17. Exploratory Factor Analysis on JS1_WorkEnv – JS12_CareerGrowth

SPSS Steps

- Analyze → Dimension Reduction → Factor
- Variables: JS1_WorkEnv through JS12_CareerGrowth
- Descriptives → check KMO and Bartlett's Test of Sphericity → Continue
- Extraction → Method: Principal Components; Based on Eigenvalue > 1 → Continue
- Rotation → Varimax → Continue → OK

Sampling Adequacy

KMO Measure of Sampling Adequacy	Bartlett's χ^2	df	p
.828	1198.91	66	< .001

KMO = .828 (“meritorious,” well above the .60 minimum), and Bartlett's test was highly significant, $\chi^2(66) = 1198.91$, $p < .001$, confirming that the correlation matrix is suitable for factor analysis.

Total Variance Explained (Eigenvalues ≥ 1)

Factor	Eigenvalue	% of Variance	Cumulative %
1	4.011	23.2	23.2
2	2.612	20.0	43.1
3	2.038	19.8	62.9

Three components had eigenvalues greater than 1 (all remaining eigenvalues were 0.535 or lower), so a 3-factor solution was retained, together explaining 62.9% of the total variance.

Rotated Component Matrix (Varimax)

Item	Factor 1	Factor 2	Factor 3
JS1_WorkEnv	.794	.184	-.010
JS2_WorkEnv	.822	.069	.069
JS3_WorkEnv	.812	.164	.032
JS4_WorkEnv	.853	.156	.091
JS5_Compensation	.087	.754	.052
JS6_Compensation	.187	.745	.035
JS7_Compensation	.108	.760	-.003
JS8_Compensation	.157	.765	.157
JS9_CareerGrowth	.067	-.034	.753
JS10_CareerGrowth	.013	.071	.793
JS11_CareerGrowth	.043	.063	.787
JS12_CareerGrowth	.033	.115	.719

APA-Style Interpretation

Principal components analysis with varimax rotation was conducted on the 12 job-satisfaction items. The Kaiser-Meyer-Olkin measure verified excellent sampling adequacy, KMO = .828, and Bartlett's test of sphericity was significant, $\chi^2(66) = 1198.91$, $p < .001$, indicating the items were sufficiently correlated for factor analysis. Three components with eigenvalues greater than 1 (4.01, 2.61, 2.04) were extracted,

together explaining 62.9% of the total variance. The rotated solution produced a clean, simple structure that mapped exactly onto the three a priori item groupings: Factor 1 (“Work Environment,” loadings .79–.85) comprised JS1–JS4; Factor 2 (“Compensation,” loadings .75–.77) comprised JS5–JS8; and Factor 3 (“Career Growth,” loadings .72–.79) comprised JS9–JS12. Each item loaded strongly ($\geq .72$) on its intended factor and negligibly ($< .19$) on the other two, supporting the construct validity of the three-subscale structure of the job satisfaction questionnaire.

Q18. Reliability Analysis: Cronbach's Alpha for the Overall Scale and Three Subscales

SPSS Steps

- Analyze → Scale → Reliability Analysis
- Run 1 – Items: all 12 items (JS1_WorkEnv – JS12_CareerGrowth); Model: Alpha → OK
- Run 2 – Items: JS1_WorkEnv – JS4_WorkEnv (Work Environment subscale) → OK
- Run 3 – Items: JS5_Compensation – JS8_Compensation (Compensation subscale) → OK
- Run 4 – Items: JS9_CareerGrowth – JS12_CareerGrowth (Career Growth subscale) → OK

Output – Reliability Statistics

Scale	N of Items	Cronbach's α	95% CI
Overall Job Satisfaction (all 12 items)	12	.813	[.772, .849]
Work Environment (JS1–JS4)	4	.900	[.876, .921]
Compensation (JS5–JS8)	4	.852	[.816, .883]
Career Growth (JS9–JS12)	4	.849	[.812, .881]

APA-Style Interpretation

The overall 12-item Job Satisfaction scale demonstrated good internal consistency, Cronbach's $\alpha = .813$, 95% CI [.772, .849]. Each of the three subscales identified in the factor analysis (Q17) also showed good-to-excellent reliability: Work Environment ($\alpha = .900$) showed excellent internal consistency, while Compensation ($\alpha = .852$) and Career Growth ($\alpha = .849$) both showed good internal consistency. All four alpha coefficients exceed the conventional .70 threshold for acceptable reliability, indicating that both the overall scale and its three subscales measure their intended constructs consistently and can be used with confidence for further analysis (e.g., as the Overall_Job_Satisfaction composite used in the logistic regression in the companion categorical-variables assignment).